

The Program in Action

When you run *hungryrobots.py*, the output will look like this:

Hungry Robots, by Al Sweigart al@inventwithpython.com

--snip--

A 20x20 grid of characters representing a noisy image of the letter 'R'. The grid contains various symbols including 'R', 'X', '@', and 'X' scattered across a background of dots and noise.

(T)eleports remaining: 2

(Q) (W) ()

(A) (S) (D)

```
Enter move or QUIT: (Z) (X) ( )
```

--snip--

How It Works

The x- and y- Cartesian coordinates that represent positions in this game allow us to use math to determine the direction in which the robots should move. In programming, x-coordinates increase going right, and y-coordinates increase going down. This means that if the robot's x-coordinate is larger than the player's coordinate, it should move left (that is, the code should subtract from its current x-coordinate) to move closer to the player. If the robot's x-coordinate is smaller, it should move right (that is, the code should add to its current x-coordinate) instead. The same applies to moving up and down based on their relative y-coordinates.

- ```
1. """Hungry Robots, by Al Sweigart al@inventwithpython.com
2. Escape the hungry robots by making them crash into each other.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: large, game"""
5.
6. import random, sys
7.
8. # Set up the constants:
```